

13. (a) Potassium dichromate(VI) is used as a reagent in acid solution in a wide range of redox reactions. The dichromate(VI) ions can be reduced using **excess** metallic zinc in acid conditions.

- Elfed performs the experiment in a sealed tube under an atmosphere of pure nitrogen gas
- Fatima performs the experiment in a standard test tube

The final products are different in each case. Use the standard electrode potentials below to identify the products in both cases. Give reasons for your answers. [4]

	Standard electrode potential, $E^\ominus/V$
$\text{Cr}^{3+}(\text{aq}) + 3\text{e}^- \rightleftharpoons \text{Cr}(\text{s})$	-0.78
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Zn}(\text{s})$	-0.76
$\text{Cr}^{3+}(\text{aq}) + \text{e}^- \rightleftharpoons \text{Cr}^{2+}(\text{aq})$	-0.42
$\text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^- \rightleftharpoons 4\text{OH}^-(\text{aq})$	+0.40
$\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+(\text{aq}) + 6\text{e}^- \rightleftharpoons 2\text{Cr}^{3+}(\text{aq}) + 7\text{H}_2\text{O}(\text{l})$	+1.33

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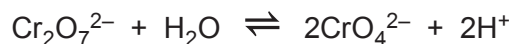
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- (b) Chromium(VI) is present in the anions dichromate(VI), $\text{Cr}_2\text{O}_7^{2-}$, and chromate(VI), CrO_4^{2-} . These ions can be interconverted in solution with the following equilibrium being established.



The equilibrium constant, K_c , for this reaction is $2.42 \times 10^{-15} \text{ mol}^2 \text{ dm}^{-6}$.

In general, dichromate(VI) is the main species in acid solution and chromate(VI) is the main species in alkaline solution.

- (i) Write an expression for the equilibrium constant, K_c , for this reaction. [1]
- (ii) The mass of the water in 1.00 cm^3 of a dilute aqueous solution is 1.00 g . Show that the concentration of water in any dilute aqueous solution is 55.5 mol dm^{-3} . [2]



- (iii) A solution was prepared containing equal concentrations of $\text{Cr}_2\text{O}_7^{2-}$ and CrO_4^{2-} . The concentration of each was $0.057 \text{ mol dm}^{-3}$. A student predicted that a solution containing equal amounts of both must be neutral. Is the student correct? Use calculation(s) to justify your answer. [4]

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- (iv) The value of the equilibrium constant for this reaction increases when the mixture is warmed. State and explain what information this provides about the enthalpy change of the reaction. [2]

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END OF PAPER

